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Title: Personalized Recommendation Workflow(Member-2)

Objective

The objective of this project is to design a workflow that can recommend suitable outfits to users based on their personal preferences and body measurements. The system aims to provide personalized suggestions that match the user's style, fit, and occasion requirements.

Workflow of the Recommendation System

Step 1: User Registration

The process starts when a new user creates an account on the platform. During registration, basic information such as name, age, and gender is collected. This information helps the system create a user profile.

Step 2: Preference Collection

After registration, the user is asked to provide fashion preferences. These preferences may include favourite colours, preferred clothing styles, preferred brands, occasions, and budget range. This information helps the system understand the user's fashion interests.

Step 3: Body Measurement Collection

To provide better-fitting outfit recommendations, the system collects body measurements such as height, weight, and body shape. These measurements help in identifying clothing items that are more suitable for the user.

Step 4: Data Processing

Once the user information is collected, the data is processed and organized. Any unnecessary or incomplete information is handled, and the data is prepared for analysis by the recommendation engine.

Step 5: Recommendation Engine

The recommendation engine analyzes the collected user data. It compares the user's preferences and body measurements with available fashion products. Based on this analysis, the system identifies outfits that closely match the user's requirements.

Step 6: Outfit Ranking

The recommended outfits are ranked according to their relevance and suitability. Factors such as style match, color preference, body fit, and popularity are considered while ranking the outfits.

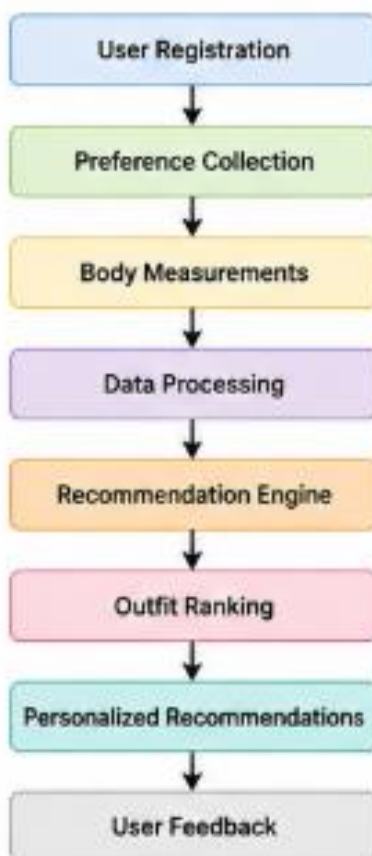
Step 7: Personalized Recommendations

The top-ranked outfits are displayed to the user as personalized recommendations. The user can view and select the outfits that best match their preferences.

Step 8: Feedback Collection

The system may collect feedback from the user regarding the recommendations. This feedback helps improve future recommendations and enhances the overall accuracy of the system.

Workflow Diagram-



Recommendation Architecture

The recommendation architecture consists of several modules working together.

User Module

Responsible for user registration, login, and profile management.

Preference Collection Module

Stores information related to fashion choices, colors, styles, and budget.

Body Measurement Module

Maintains body-related information for better outfit fitting.

Data Processing Module

Processes and organizes user information before recommendation generation.

Recommendation Engine

Uses recommendation algorithms to identify suitable outfits.

Outfit Ranking Module

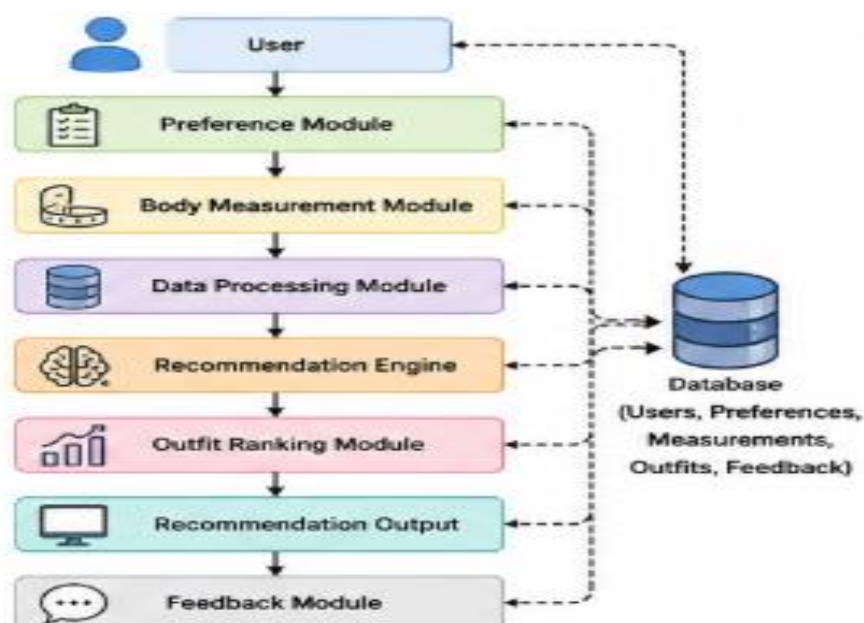
Ranks outfit suggestions according to relevance and user preferences.

Recommendation Output Module

Displays the final personalized outfit recommendations.

Feedback Module

Collects user feedback and improves future recommendations.



Expected Outputs

The system is expected to provide:

- Personalized outfit recommendations
- Better-fitting clothing suggestions
- Outfit suggestions for different occasions
- Ranked outfit lists based on user preferences
- Improved recommendations through feedback

UI Mock-up Example-

The UI mock-up consists of two main sections. The 'Your Preferences' section on the left is a light blue box containing several input fields: 'Favorite Color' (dropdown menu set to 'Black'), 'Preferred Style' (dropdown menu set to 'Casual'), 'Occasion' (dropdown menu set to 'College'), 'Budget' (dropdown menu set to '₹2000'), 'Height' (text input set to '165 cm'), and 'Weight' (text input set to '58 kg'). Below these fields is a blue button labeled 'Get Recommendations'. An arrow points from this section to the 'Recommended Outfits' section on the right. This section is a light green box containing three outfit cards. Each card shows a mannequin wearing a different outfit, labeled 'Outfit 1', 'Outfit 2', and 'Outfit 3' respectively. Below each outfit image are five yellow stars, indicating a rating.

Recommendation Techniques Used

Method	Description
Content-Based Filtering	Recommends items similar to user preferences
Collaborative Filtering	Uses preferences of similar users
Hybrid Recommendation	Combines both methods for better accuracy

The Personalized Outfit Recommendation Workflow provides a systematic approach for generating customized fashion suggestions. By combining user preferences, body measurements, and recommendation techniques, the system can help users discover outfits that match their style and fit requirements. This improves user satisfaction and enhances the online shopping experience.